Atul Marichetty

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EDUCATION

Rutgers University–New Brunswick **Expected May 2028**
Bachelor of Science in Computer Science and Mathematics **(GPA: 3.93/4.0)** *New Brunswick, NJ*

- **Relevant Coursework:** Data Structures, Design and Analysis of Computer Algorithms, Computer Architecture, Mathematical Theory of Probability, Calculus I-IV, Mathematical Reasoning, Linear Algebra, Statistics
- **Activities:** Kappa Theta Pi Professional Fraternity (Technology Committee), Undergraduate Student Alliance of Computer Scientists Mentor, Rutgers Quantitative Finance Club Member, Academic Tutor (CS, Math, Science)

EXPERIENCE

Fanatics **Jun. 2026 – Aug. 2026**
Software Engineer Intern *New York, NY*

- Built and deployed a multimodal RAG financial assistant (Azure AI Search, PyMuPDF vision, FastAPI, Docker, Redis) with tiered model routing that injects validated financial tables per query, cutting manual lookup by 6 hrs/week for 20+ employees
- Engineered a validation-first pipeline converting a 228-page financial binder into 139 tables verified by 230+ arithmetic tie-outs
- Reduced M&A team manual processing time by 50% by integrating agentic AI configurations (Power Automate, Claude Skills, Airtable, Copilot) to automate critical workflow processes such as data parsing, schema mapping, and risk analysis

ServiceAgent **Sept. 2025 – Dec. 2025**
AI Engineer Intern *New Brunswick, NJ*

- Engineered an AI workflow using n8n, Excel, and OpenAI API to process unstructured CRM notes, perform semantic analysis, extract insights, and generate sales recommendations, boosting Business Development team call bookings by 50%
- Built a full-stack FastAPI/React platform for AI-driven candidate sourcing and interviewing for staffing and technology clients

Headstarter **Dec. 2024 – Feb. 2025**
Software Engineer Intern *Remote*

- Benchmarked Logistic Regression, Random Forest, and KNN on 1,100+ weeks of market data to predict market anomalies (90% accuracy, 0.79 precision), and deployed best-performing model as Streamlit app with LLM-generated explanations
- Developed a brain tumor classification model using transfer learning (Xception CNN) in TensorFlow/Keras, achieving 92% test accuracy across 4 tumor classes on 7,000+ MRI scans; engineered a preprocessing pipeline with data augmentation

PROJECTS

MoneyMentor | ***React, TypeScript, Flask, SQLite, GPT***

- Built a full-stack personal finance dashboard using React (TypeScript), Flask, and SQLite, featuring CSV bank statement ingestion, rule-based transaction categorization, and interactive spending visualizations with Tailwind CSS
- Designed a RESTful API with 10+ endpoints for spending summaries, category breakdowns, income tracking, and net balances
- Integrated a GPT-powered financial advisor chatbot supporting multi-turn conversations, CSV-based financial context injection, and session management with fallback support for OpenAI-compatible API endpoints

Breast Cancer Detector | ***Python, scikit-learn, pandas, NumPy, Matplotlib***

- Engineered an end-to-end ML classification pipeline over 569 tumor samples and 30 diagnostic features, benchmarking 6 classifiers across 3 feature sets (18 model configurations) to distinguish malignant from benign tumors
- Optimized hyperparameters via 5-fold stratified GridSearchCV tuned for recall, reaching 93% malignant-tumor recall while applying PCA and ANOVA F-score selection to reduce dimensionality from 30 to 10–15 features

AI Document Analyzer | ***Python, Streamlit, OpenAI, FAISS, Pydantic***

- Built a document intelligence app with a Streamlit UI that extracts text from uploaded PDFs and uses GPT with 5 document-type-specific prompts to return JSON summaries, key entities, financial figures, action items, and risks
- Engineered a RAG pipeline using FAISS similarity search over OpenAI embeddings, with overlapping 800-character chunking and top-5 retrieval, enabling truncation-free analysis of 12K+ character documents and a document-grounded Q&A chatbot
- Hardened the LLM pipeline for reliability with Pydantic schema validation and automatic coercion of malformed outputs, 3-attempt retry logic for invalid JSON and API failures, and graceful error handling for scanned, empty, or unreadable PDFs

TECHNICAL SKILLS

Programming Languages: Python, Java, HTML/CSS, JavaScript, TypeScript, SQL, Swift, C++
Tools: Docker, Redis, Git, Azure, Jira, Confluence, GitHub, AWS, VS Code, Eclipse, Xcode, Tableau, PostgreSQL, Claude, Codex
Frameworks & Libraries: FastAPI, React, Node.js, Flask, scikit-learn, pandas, TensorFlow, Keras, NumPy, Next.js